

Simple Low-cost Intercom

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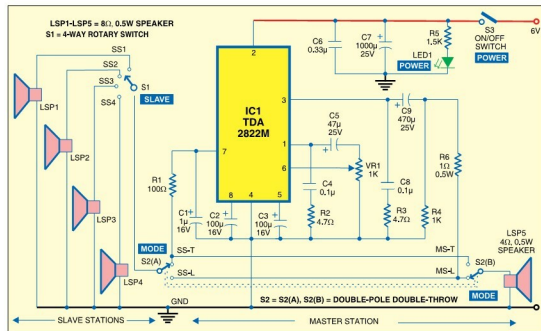
Sometimes you need to communicate from your room with someone in another room of the house, in the garage, in the courtyard or at the entrance door. Or, you may need to monitor your children or an elderly relative in the other room. In all these situations, this simple audio intercom comes handy. It is built around

speaker (working as a microphone) is amplified by the first amplifier in TDA2822M. The output level is regulated using potentiometer VR1. If regulation is not needed, replace potentiometer VR1 with a resistor having a value of 100 ohms to 1 kilo-ohm. The output of the first amplifier is further amplified by the second channel of the integrated circuit TDA2822 and fed to the selected loudspeaker. Both

and switch S2(A) is in position SS-T, the signal from the loudspeaker of the master station (LSP5) is amplified by the IC, adjusted using VR1 and fed to the loudspeaker of the master station (LSP5).

Similarly, if switch S2(B) is in position MS-T and switch S2(A) is in the position SS-L, the signal from the loudspeaker of the master station (LSP5) is amplified by the IC, adjusted using VR1 and sent to the loudspeaker of the slave station (LSP1). Switch S3 is the on/off switch of the intercom. The master station can do audio surveillance with any of the four speakers of the slave stations.

The circuit works off a 6V rechargeable battery, or you can use a 6V, 500mA adaptor. As



the popular low-cost stereo amplifier TDA2822M.

The circuit consists of four slave stations (SS1 through SS4), each having a loudspeaker. The loudspeakers LSP1, LSP2, LSP3 and LSP4 in slave stations are selected by rotating switch S1. It is important that loudspeakers LSP1 through LSP4 are small (e.g. 0.5W) and sensitive with a resistance of at least 8 ohms. Small loudspeakers from portable radios can be used for the purpose.

The intercom also has a main station (MS) which contains the circuit including switches S1 through S3, the power supply and loudspeaker LSP5.

The signal from the selected loud-

channels of TDA2822M have a gain of around 100 (40 dB), which is appropriate for the circuit.

The selection of the loudspeakers and the working mode of the intercom is made by switches S1, S2 and S3 as follows: Switch S1 is used for selecting the active slave station. For example, if S1 is in position SS1, slave station SS1 with LSP1 is selected. Switch S2 is used for selecting the operation mode for the master station. There are two modes: Master station-listen (MS-L) and master station-talking (MS-T). An example below explains the operation with slave station SS1.

If switch S2(B) is in position MS-L